

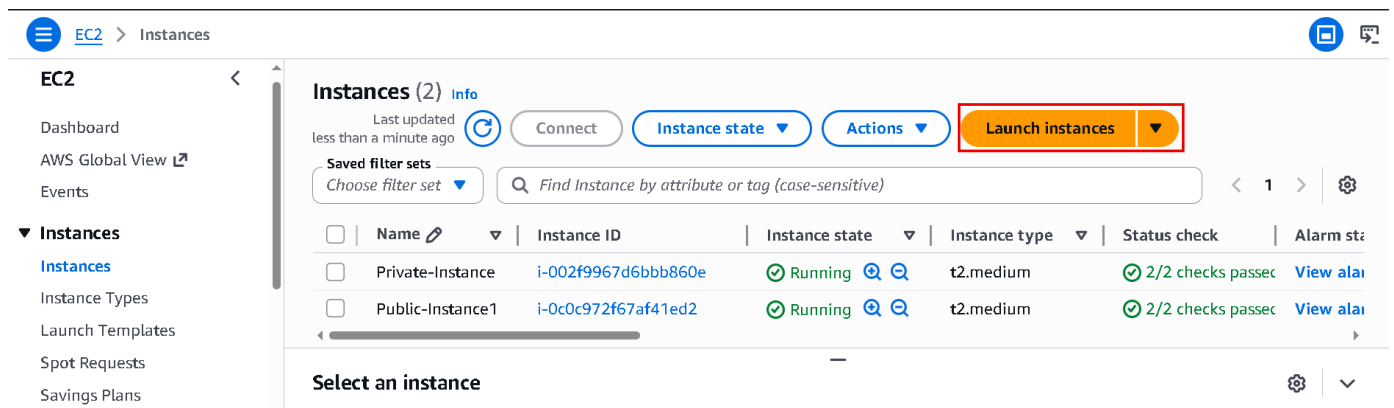
Customize Amazon Machine Image

Summary: -

This document explains how to create a custom Amazon Machine Image (AMI) from a Windows EC2 instance after installing server roles and features. It shows how to launch a Windows instance, connect through RDP, install Web Server and DHCP roles, create an AMI from the configured instance, and launch a new EC2 instance using that custom AMI so the same setup and website are already available.

Let's Create: -

- Search EC2 in Aws Console and Open it.
- Now Let's create windows EC2 instance.



- Now opens EC2 in new tab.
- And click Launch instance.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

[Add additional tags](#)

- Enter instance name.
- Scroll down.

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

 Search our full catalog including 1000s of application and OS images

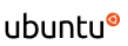
Quick Start



Amazon Linux



macOS



Ubuntu



Windows



Red Hat



SUSE



[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

- Choose OS click Browse more AMIs.

 [EC2](#) > [Instances](#) > [Launch an instance](#) > AMIs 

 Search for an AMI by entering a search term e.g. "Windows" 

Quick Start AMIs (45)

Commonly used AMIs

My AMIs (13)

Created by me

AWS Marketplace AMIs (6091)

AWS & trusted third-party AMIs

Community AMIs (500)

Published by anyone



Windows

Free tier eligible

Verified provider

Microsoft Windows Server 2019 Base

ami-08e30c01541e6a4f3 (64-bit (x86))

Microsoft Windows 2019 Datacenter edition.
[English]

Platform: windows
Virtualization: hvm

Root device type: ebs
ENA enabled: Yes

Select

 64-bit (x86)

- I am going to use windows 2019 Base.
- Click on select and check first they are Free tier eligible or not.

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

 Search our full catalog including 1000s of application and OS images

AMI from catalog

Recents

My AMIs

Quick Start

Name

Microsoft Windows Server 2019 Base

Verified provider

Free tier eligible



[Browse more AMIs](#)

Including AMIs from
AWS, Marketplace and
the Community

Description

Microsoft Windows 2019 Datacenter edition. [English]

Microsoft Windows Server 2019 with Desktop Experience Locale English AMI provided by Amazon

Image ID

ami-08e30c01541e6a4f3

- Now scroll down.

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro

Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand SUSE base pricing: 0.0104 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour

On-Demand RHEL base pricing: 0.0392 USD per Hour

On-Demand Windows base pricing: 0.0196 USD per Hour

On-Demand Linux base pricing: 0.0104 USD per Hour

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

- now we are just going to select this t3.micro
- In this we are going to get two virtual CPU and one GB Ram.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Select

 [Create new key pair](#)

- Click Create new key pair.

Create key pair

Key pair name

Key pairs allow you to connect to your instance securely.

demo

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

Cancel

Create key pair

- Enter key name.
- Key pair type as RSA
- And .pem file
- And last click Create key pair
- Now your keypair is downloaded and save it.
- Now scroll down.

▼ Network settings Info

Edit

Network Info

vpc-072c4acd761c3b942

Subnet Info

No preference (Default subnet in any availability zone)

Auto-assign public IP Info

Enable

Additional charges apply when outside of free tier allowance

- Now click Edit.

▼ Network settings [Info](#)

VPC - *required* | [Info](#)

vpc-072c4acd761c3b942
172.31.0.0/16

(default) ▼



Subnet | [Info](#)

No preference ▼



[Create new subnet](#) ↗

Availability Zone | [Info](#)

No preference ▼



[Enable additional zones](#) ↗

Auto-assign public IP | [Info](#)

Enable ▼

[Additional charges apply](#) when outside of [free tier allowance](#)

- Now Default VPC we are not changing it.
- Scroll down.

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

Security group name - *required*

launch-wizard-80

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and . _ - / 0 # , @ [] + = & ; ! \$ *

Description - *required* | [Info](#)

launch-wizard-80 created 2026-05-19T08:26:27.229Z

- Now here create a security group or you select your created security group also.
- Aws always create a custom security group if you want to change name of security group you can change it.

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 3389, 0.0.0.0/0)

Remove

Type [Info](#)

rdp

Protocol [Info](#)

TCP

Port range [Info](#)

3389

Source type [Info](#)

Anywhere

Source [Info](#)

Q Add CIDR, prefix list or security grou

0.0.0.0/0 X

Description - optional [Info](#)

e.g. SSH for admin desktop

▼ Security group rule 2 (TCP, 80, 0.0.0.0/0)

Remove

Type [Info](#)

Custom TCP

Protocol [Info](#)

TCP

Port range [Info](#)

80

Source type [Info](#)

Custom

Source [Info](#)

Q Add CIDR, prefix list or security grou

0.0.0.0/0 X

Description - optional [Info](#)

e.g. SSH for admin desktop

- You can add extra permissions in your security group.
- By default for windows rdb permissions is added.
- We add one Custom TCP, Port range 80 and source 0.0.0.0/0
- Now scroll down.

▼ **Configure storage** [Info](#) Advanced

1x GiB Root volume, Not encrypted

[Free tier eligible customers can get up to 30 GB of EBS General Purpose \(SSD\) or Magnetic storage](#)

[Add new volume](#)

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

[Click refresh to view backup information](#)
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

File systems

▼ **Summary**

Number of instances [Info](#)

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 30 GiB

[Free tier:](#) In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't

- Suppose if you are going to buy a laptop, they will ask you right that what is the storage Requirement like One TB or 512 GB etc.
- Same way here.
- It will provide you 30 GB volume.
- Now this volume is actually root volume.
- It means it will install operating system inside this volume.
- Now last thing in summary you give how much instance you need currently I need 1 so it by default 1 if I need more I increase it.
- Now simply scroll down.

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 30 GiB

Cancel

Launch instance

Preview code

- And click Launch instance.

Instances (1/4) [Info](#)

Last updated 1 minute ago

Connect

Instance state ▼

Actions ▼

Launch instances ▼

Saved filter sets

Choose filter set ▼

Find Instance by attribute or tag (case-sensitive)

< 1 >

⚙️

☐	Name 🔗 ▼	Instance ID	Instance state ▼	Instance type ▼	Status check	Availability Zone ▼	Public IPv4
☒	MyServer_AMI	i-0b15d14555dbdb678	Running 🔍	t3.micro	Initializing 🕒	ap-south-1a	ec2-13-206-196-194

i-0b15d14555dbdb678 (MyServer_AMI)

⚙️ ▼

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Instance summary [Info](#)

Instance ID

[🔗](#) i-0b15d14555dbdb678

Public IPv4 address

[🔗](#) 13.206.196.194 | [open address](#) [🔗](#)

Private IPv4 addresses

[🔗](#) 172.31.44.54

- Now see our status check becomes 3/3 checks passed.
- Now connect with our Machine.
- Select the created instance click on connect.

Connect [Info](#)

Connect to an instance using the browser-based client.

Instance ID

[🔗](#) i-0b15d14555dbdb678 (MyServer_AMI)

VPC ID

[🔗](#) vpc-072c4acd761c3b942

Security groups

[🔗](#) sg-070fb510f1699db0f (launch-wizard-203)

IAM role

—

SSM Session Manager

RDP client

EC2 serial console

- Now click RDP client.
- And scroll down.

Instance ID

 i-Ob15d14555dbdb678 (MyServer_AMI)

Connection Type

- ☒ Connect using RDP client
Download a file to use with your RDP client and retrieve your password.

☐ Connect using Fleet Manager
To connect to the instance using Fleet Manager Remote Desktop, the SSM Agent must be installed and running on the instance. For more information, see [Working with SSM Agent](#) 

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

 [Download remote desktop file](#)

When prompted, connect to your instance using the following username and password:

Public DNS

 ec2-13-206-196-194.ap-south-1.compute.amazonaws.com

Username [Info](#)

 Administrator 

Password [Get password](#)

- Now click on download remote desktop file.
- When your file is downloaded go to download folder and open it.

Make sure that you trust the remote computer before you connect.

This remote connection could harm your local or remote computer.

PC Name: ec2-13-235-18-93.ap-south-1.compute.amaz

Let the remote computer access the following resources:

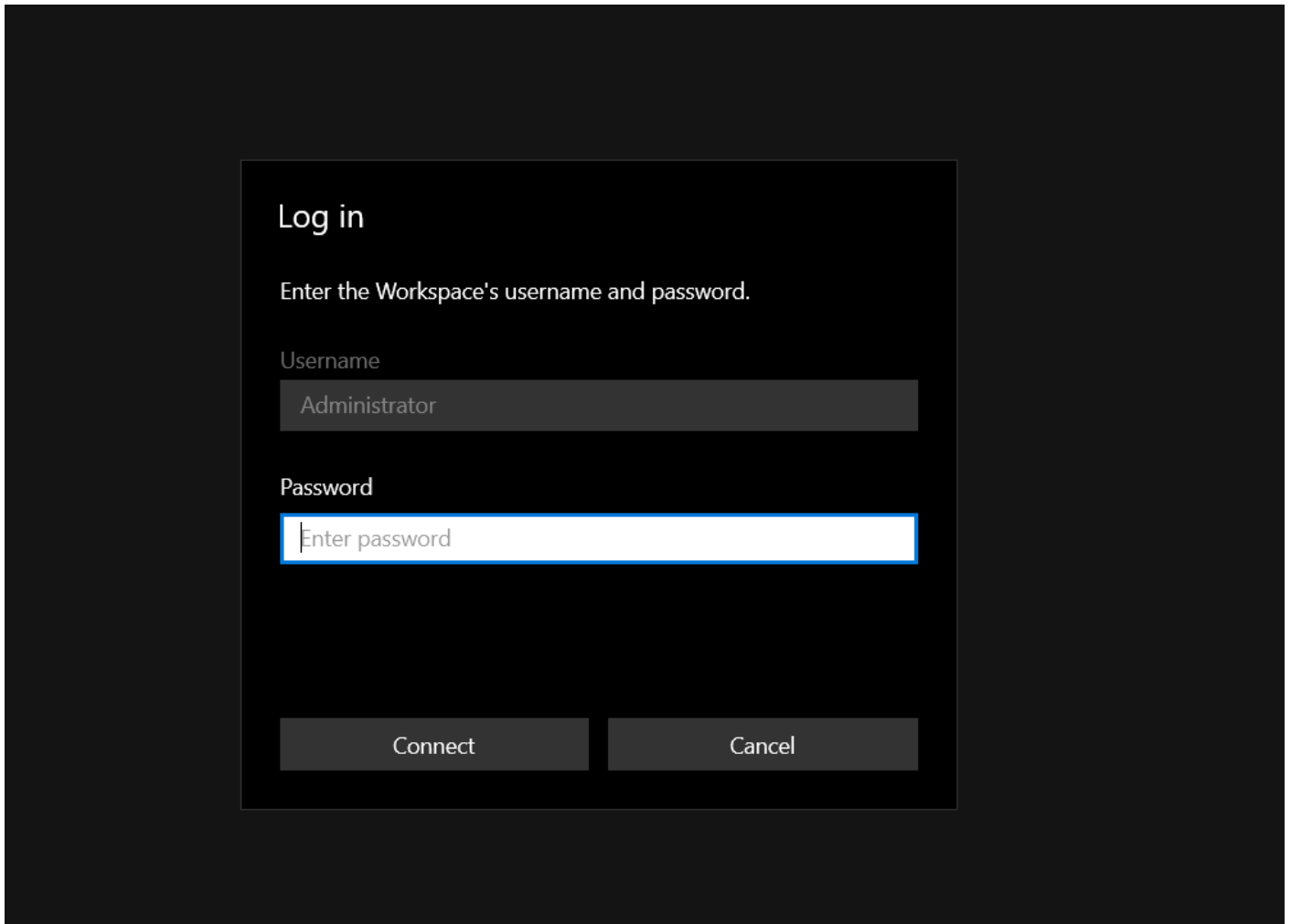
☒ Clipboard

☐ Don't ask me again

Continue

Cancel

- Now click continue.



- Now we need a password.
- Remember you created a demo.pem key in starting now we use it.
- Go to your browser instance tab.

Instance ID

 i-0b15d14555dbdb678 (MyServer_AMI)

Connection Type

- ☒ **Connect using RDP client**
Download a file to use with your RDP client and retrieve your password.

- ☐ **Connect using Fleet Manager**
To connect to the instance using Fleet Manager Remote Desktop, the SSM Agent must be installed and running on the instance. For more information, see [Working with SSM Agent](#) 

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

 [Download remote desktop file](#)

When prompted, connect to your instance using the following username and password:

Public DNS

 ec2-13-206-196-194.ap-south-1.compute.amazonaws.com

Username [Info](#)

 Administrator 

Password [Get password](#)

- Now click on get password.

Get Windows password [Info](#)

Use your private key to retrieve and decrypt the initial Windows administrator password for this instance.

Instance ID

 i-Ob15d14555dbdb678 (MyServer_AMI)

Key pair associated with this instance

 demo

Private key

Either upload your private key file or copy and paste its contents into the field below.

 Upload private key file

Private key contents

- Now click on upload private key file.
- So you upload your demo.pem file.

 Upload private key file

demo.pem
1.67 KB



Private key contents

```
-----BEGIN RSA PRIVATE KEY-----
MIIEowIBAAKCAQEAvv236ItS1zMgilPBjRBrK5qInIJ5LXDmX9bdpc6O+jrL5+t2
HcGPhXN/Doy3yaFD2Tm5N8OUpcNv6vh4LWBP7RBXe59bQgLepILZ+BsB+fQ02oAw
kfeIEBxHC+5Rpz8Dm92rsOJxhfw/8+75K1wD4ywRhrtSRKcqZu4LxsWmE2HCoEhZ
eWn8yE/mmgCuiV1XjsfthocEBvcy1RK63RB+F58SyyXa7XudRzqZM46k9xncurFv
MxKfZJHT+e/diwVDrAH/Ggt2+yep4U5LxqUK2WrusG0PnkOFFxAn8P6igS+Jr5ru
KS7S0veZ/pROglhmlhLO2qUL46VHXzMnGjFlgQwIDAQABAolBAGFvWpZKxLZYtO8c
z49B0wr5qVv/DAbso/qcyTK/cB2kRUYPz3Zauvdf9p/blh9dEBoP4rfaJULzrPL4
```

Cancel

Decrypt password

- Now simply scroll and click Decrypt password.

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

 Download remote desktop file

When prompted, connect to your instance using the following username and password:

Public DNS

 ec2-13-235-18-93.ap-south-1.compute.amazonaws.com

Username [Info](#)

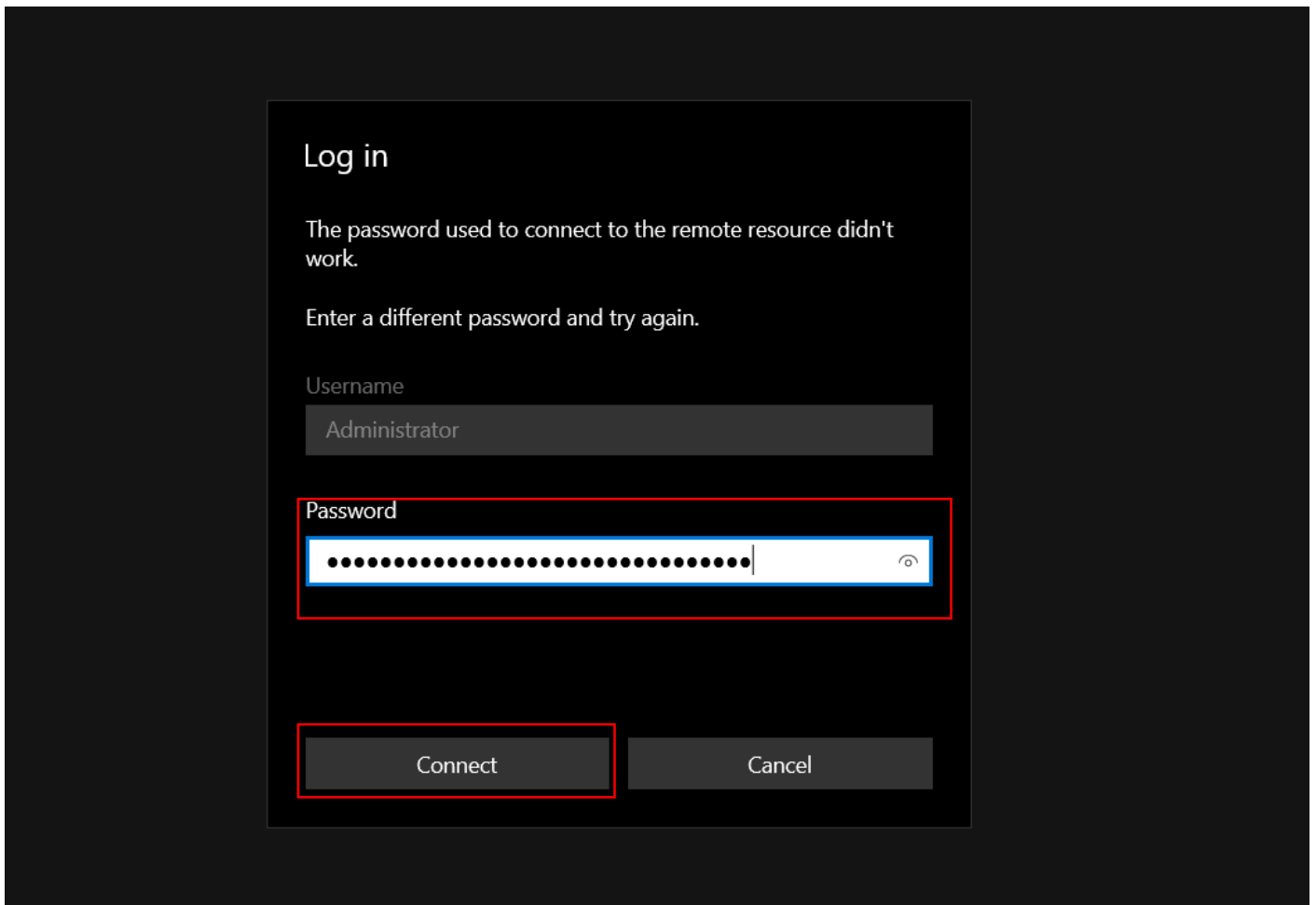
 Administrator

Password

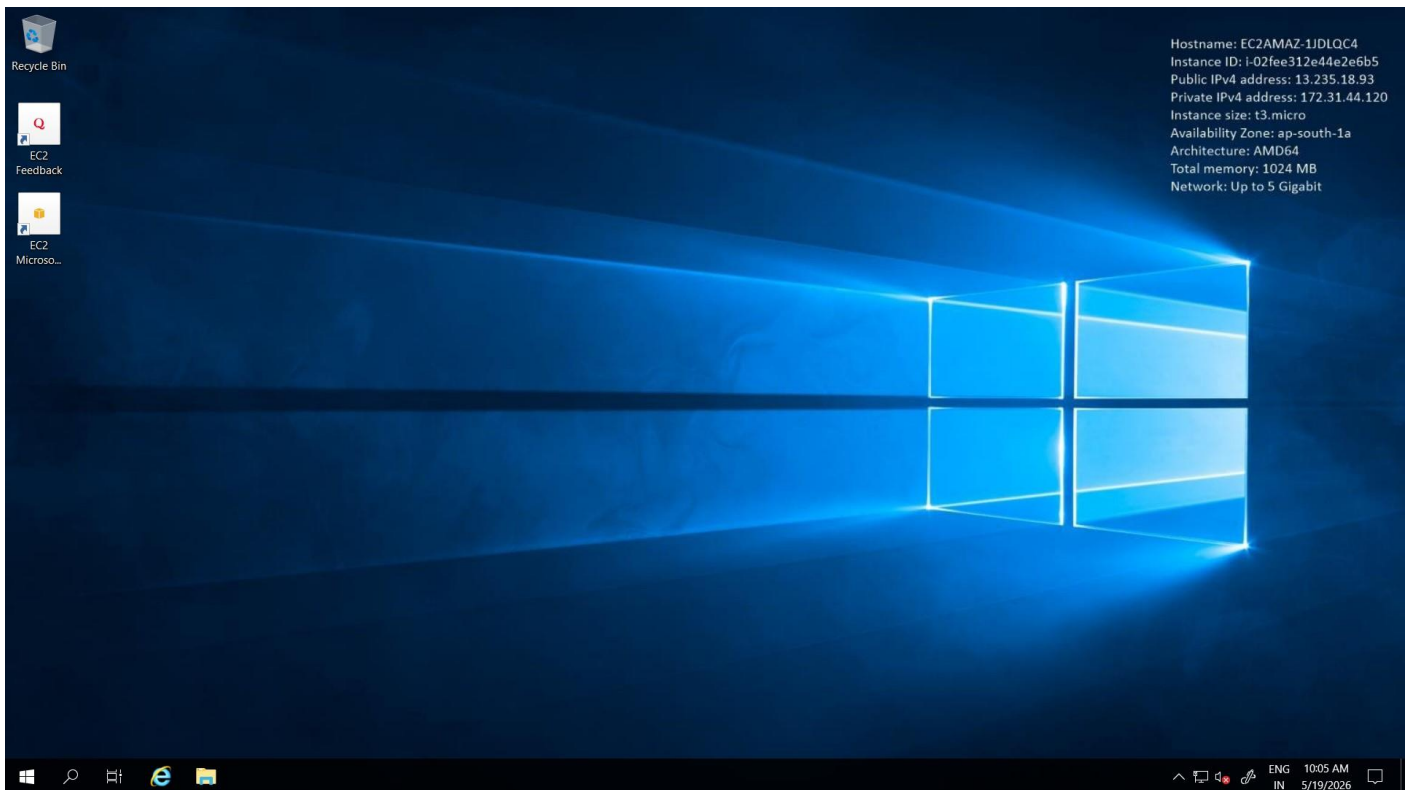
 @JNfy?%;z=wJhuaaavg;!%ukFy;gi8Dv

 If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.

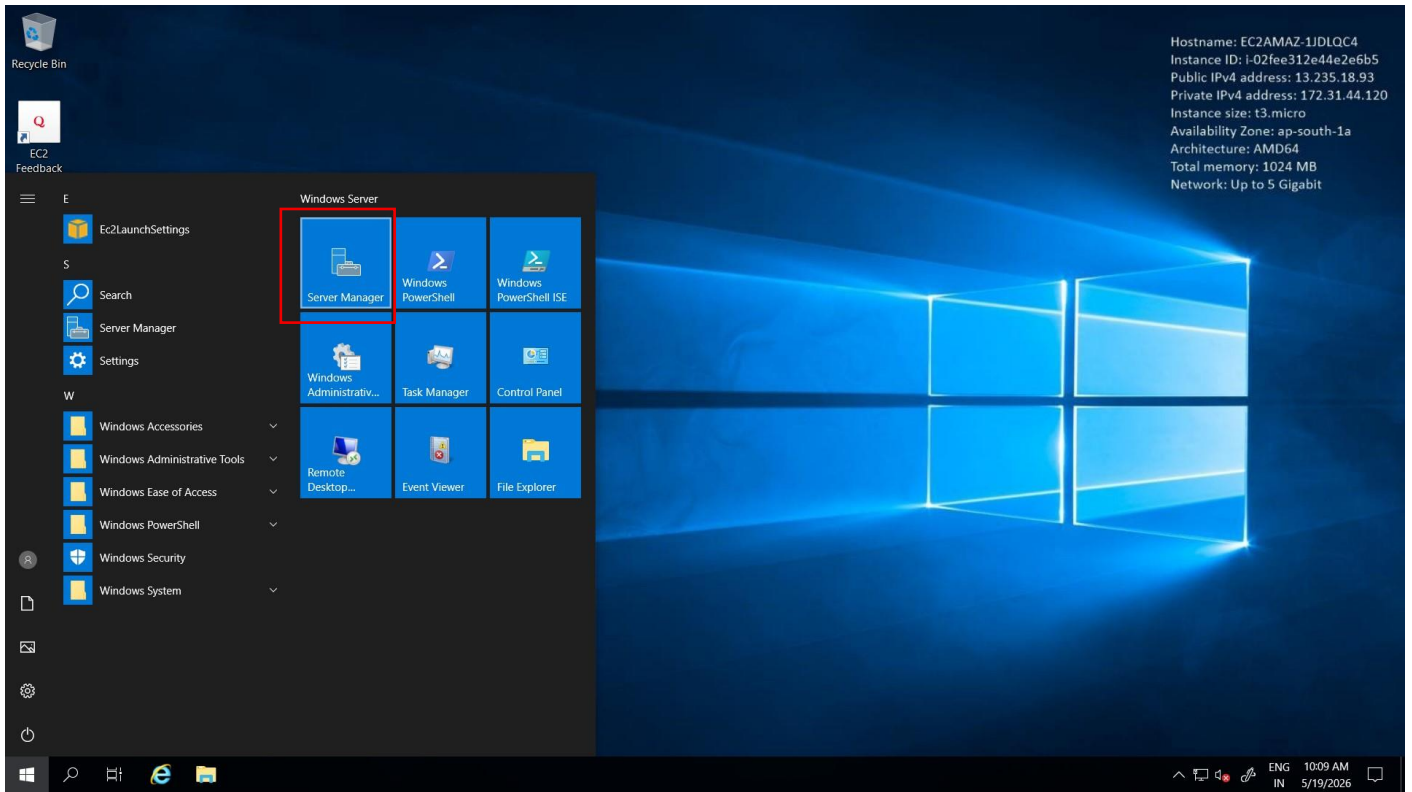
- Now simply copy the Password.



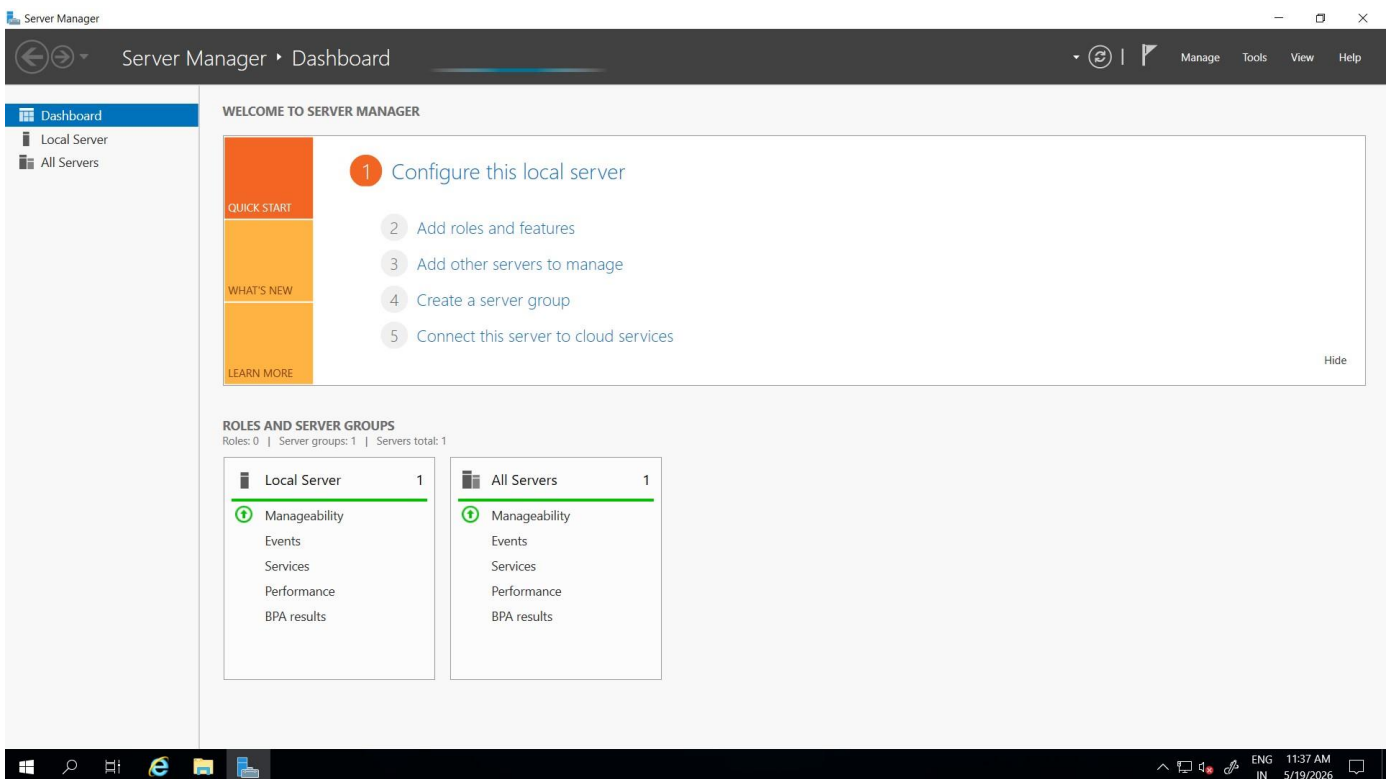
- Now paste the password in remote Desktop and click connect.



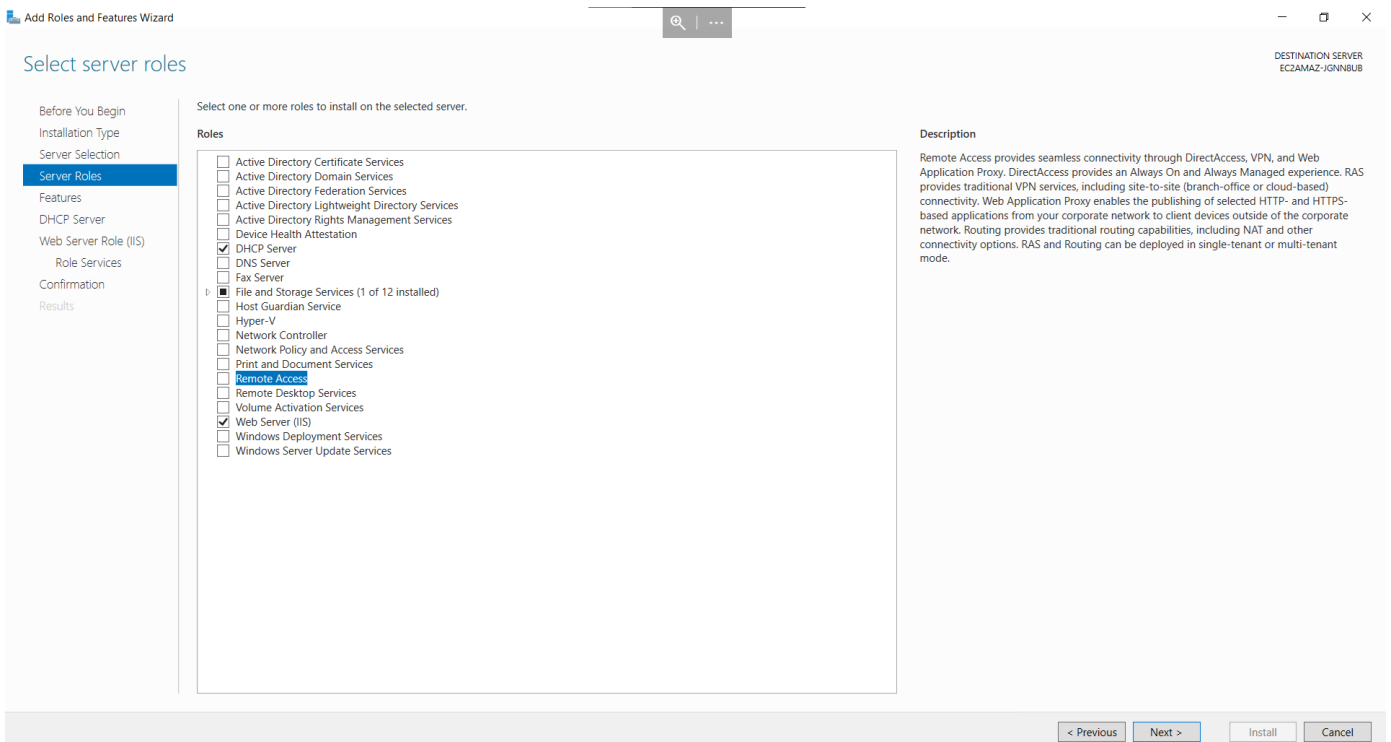
- Now you see this is the window machine currently we created.
- Now we want to change windows password.
- Click on windows icon in machine.



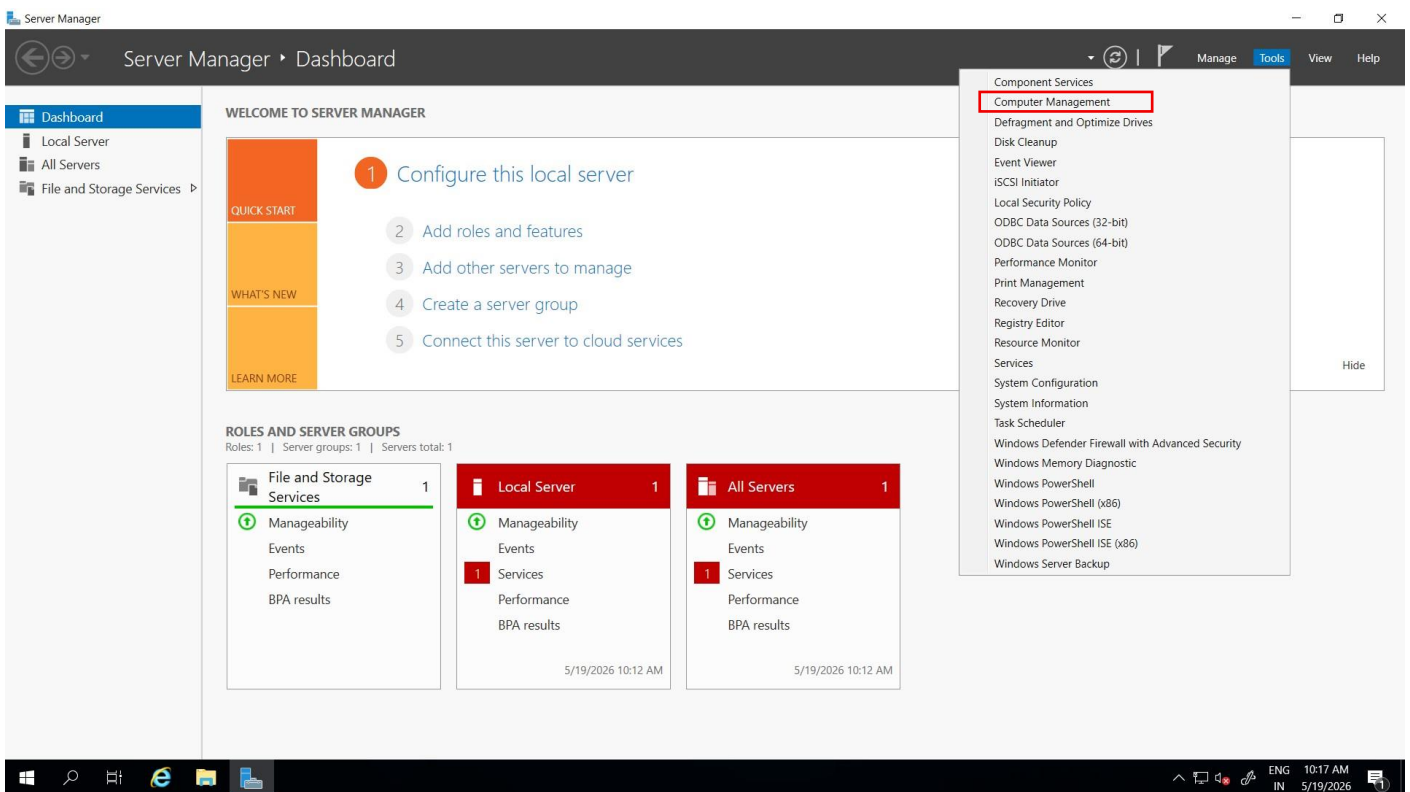
- Now click server manager.



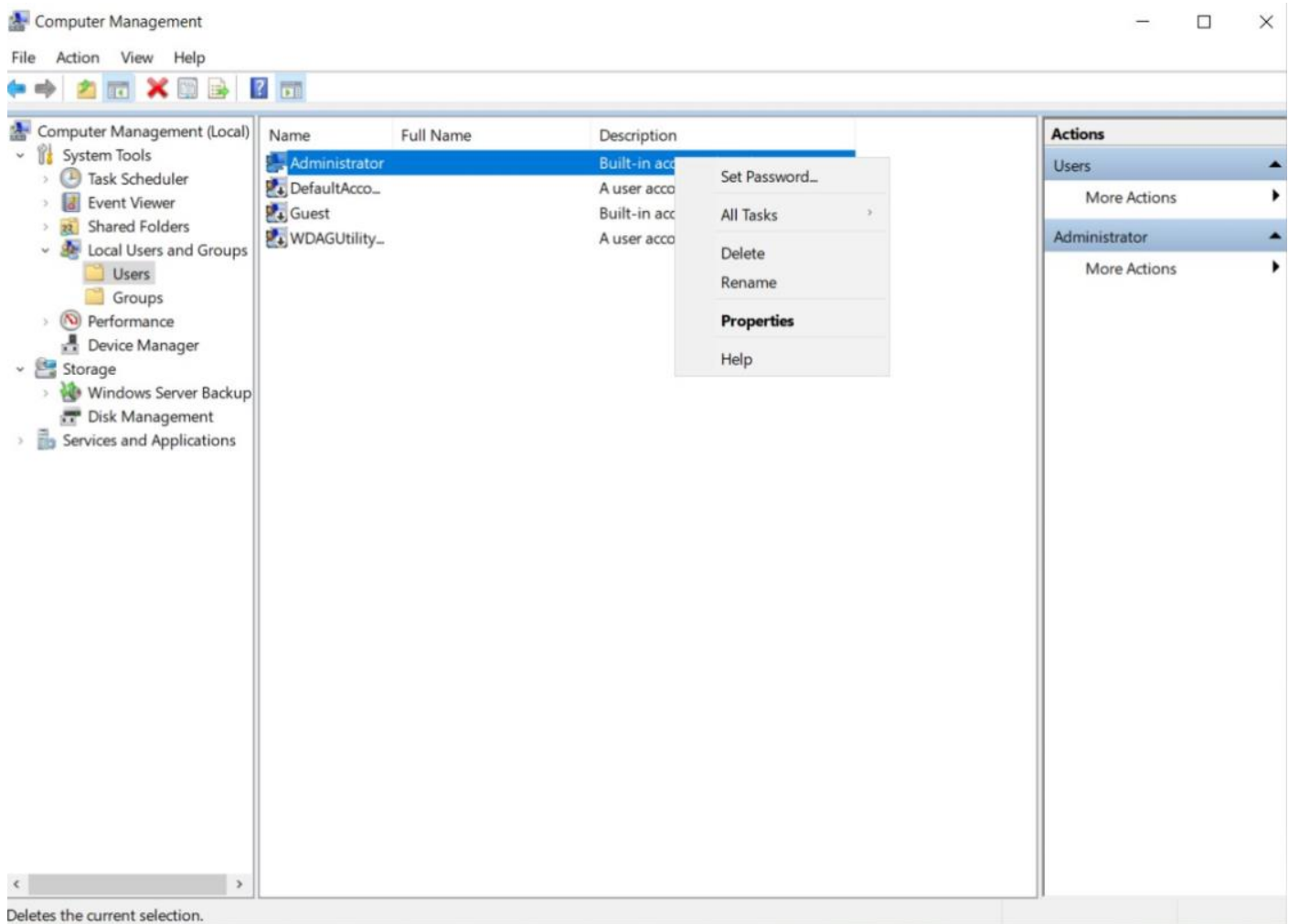
- Now select ADD roles and Features.



- Now in server Roles select DHCP server and Web Server
- And in every stage only click Next and last stage click install.
- When its done.



- Now click Tool and then select Computer management.



- now click Local user and group and then click user.
- Now on Administrator right click and select set password.
- Then simply Enter new password.
- So this is for when you again open this you can use your created password for login in this machine.
- Now go back to your main screen and go inside browser and open Ec2 instance tab

Instances (1/4) Info Last updated 26 minutes ago Refresh Connect Instance state Actions Launch instances

Saved filter sets
Choose filter set

Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability
MyServer_AMI	i-0b15d14555dbdb678	Running	t3.micro	Initializing	ap-south-

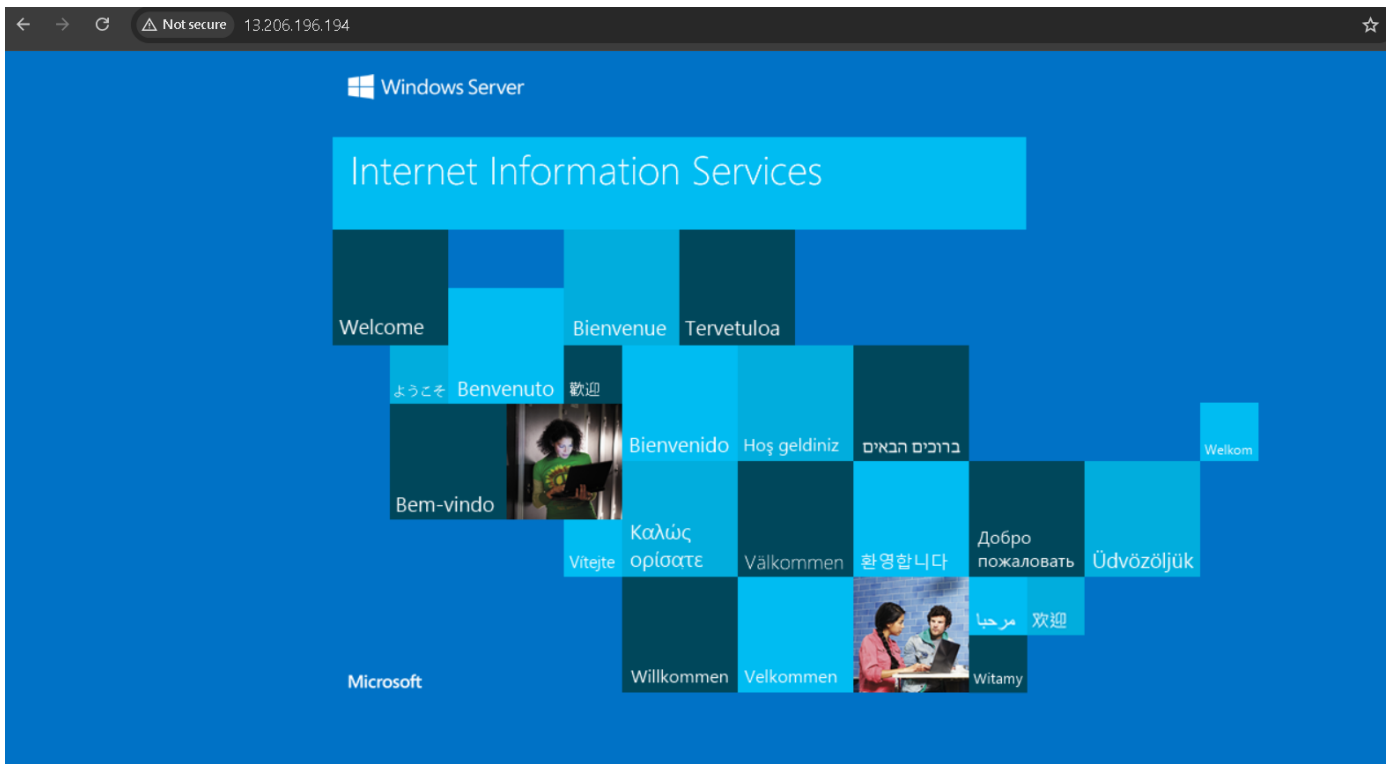
i-0b15d14555dbdb678 (MyServer_AMI)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

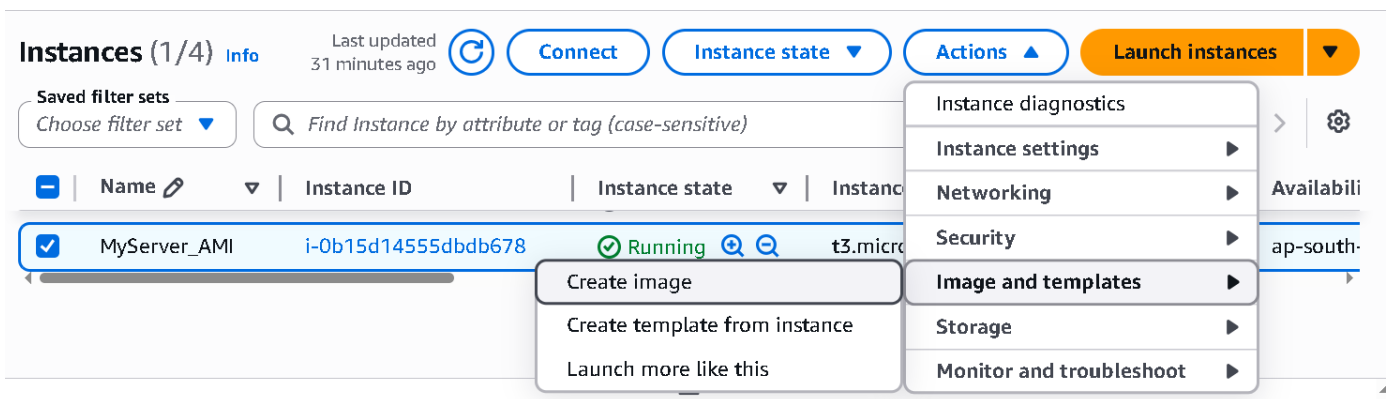
▼ Instance summary Info

Instance ID i-0b15d14555dbdb678	Public IPv4 address 13.206.196.194 open address	Private IPv4 addresses 172.31.44.54
------------------------------------	--	--

- Copy IPv4 Address.
- And open new tab and paste it and check its working or not.



- Now its working means our features and roles is added correctly.
- Now we make it a image.
- Go back to Ec2 tab.



- Select your instance and click Image and template.
- And then click Create image.

Image details

Instance ID

 [i-0b15d14555dbdb678](#) (MyServer_AMI)

Image name

Maximum 127 characters. Can't be modified after creation.

Image description - optional

Maximum 255 characters

☒ Reboot instance

When selected, Amazon EC2 reboots the instance so that data is at rest when snapshots of the attached volumes are taken. This ensures data consistency.

Instance volumes

- Simply give your AMI name and scroll down and click create image.

- Now simply done and click AMI's

- Now wait when the status of your created AMI become Available.
- Then go to instance and delete created EC2 instance.
- And then create new instance.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

- Enter name.

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

- Now in Application and OS Images select My AMIs
- And then select your Custom_AMI that you created now.
- Now follow same step that you follow in creation of first instance.

Instances (1/3) [Info](#)

Last updated less than a minute ago [Refresh](#) [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets [Choose filter set](#) [<](#) [1](#) [>](#) [Settings](#)

	Name	Instance ID	Instance state	Instance type	Status check	Availability
☐	MyServer_AMI	i-0b15d14555dbdb678	Terminated	t3.micro	–	ap-south-
☒	MyServer_AMI2	i-09cdaae9c58e207eb	Running	t3.micro	Initializing	ap-south-

- Now see new instance is created and wait for status become 3/3 check passed.

Instances (1/3) [Info](#)

Last updated less than a minute ago [Refresh](#) [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets [Choose filter set](#) [<](#) [1](#) [>](#) [Settings](#)

	Name	Instance ID	Instance state	Instance type	Status check	Availability
☒	MyServer_AMI2	i-09cdaae9c58e207eb	Running	t3.micro	3/3 checks passed	ap-south-

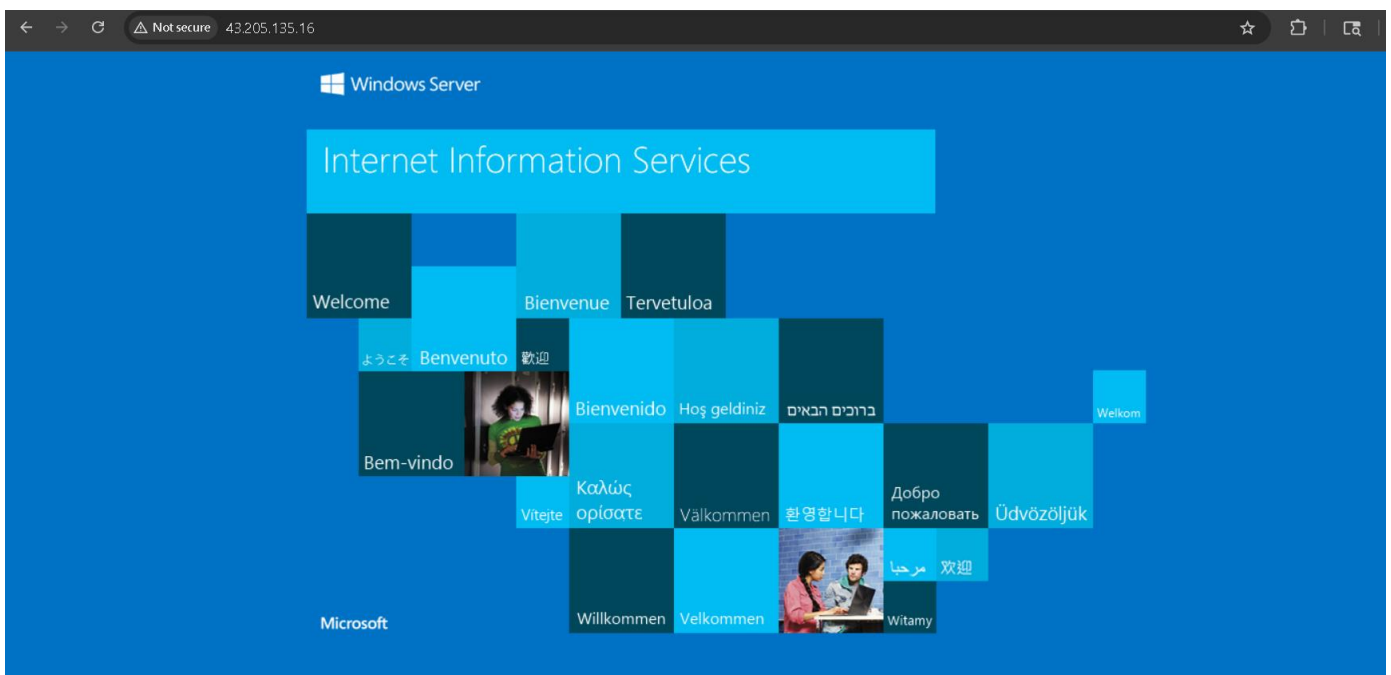
i-09cdaae9c58e207eb (MyServer_AMI2)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

▼ **Instance summary** [Info](#)

Instance ID i-09cdaae9c58e207eb	Public IPv4 address 43.205.135.16 open address	Private IPv4 addresses 172.31.33.102
---	--	--

- Now copy the IPv4 address and open in new browser tab.



- Now you see our website is open.
- Now you see we don't open our machine and add roles and features in this new EC2 instance.
- But it's working because we use image of previous instance at creation time of new instance.